

TCS IPA PREPARATION 2025

UNIX MCQs

Section 1: Multiple Choice Questions (MCQs)

1. The command used to find the count of only words in a file?

- a. wc -w <filename>
- b. wc -words <filename>
- c. wc <filename>
- d. None of the above

Correct Answer: a

2. What will be the output of given awk script? awk 'BEGIN{s=0 while (s < 55) {print s;++s}}'

- a. Prints the numbers from 0 to 55.
- b. Prints the numbers from 0 to 5d.
- c. Command will throw a syntax error
- d. —

Correct Answer: C

3. What is the command to view content of file?

- a. cat
- b. ls
- c. vi
- d. —

Correct Answer: a

4. What is command to switch to another user?

- a. su
- b. switch
- c. user
- d. login

Correct Answer: a

5. What is command to set an environment variable?

- a. env
- b. export
- c. Set
- d. Unset

Correct Answer: b

6. What is command to change root directory?

- a. chroot
- b. chmod
- c. chown
- d. chgrp

Correct Answer: a

7. What is command to run process in background?

- a. &
- b. /
- c. Ctrl + c
- d. Ctrl + v

Correct Answer: a

8. Which Symbol be used with grep command to match the pattern pat at the beginning of a line?

- a. ^pat
- b. \$pat
- c. pat\$
- d. pat^

Correct Answer: a

9. Which command is used to sort the lines of a data in a file in reverse order?

- a. sort
- b. sh
- c. sort -r
- d. st

Correct Answer: c

10. Which of the command to change the protection mode of files starting with string emp and ending with 1, 2 or 3?

- a. chmod u+x emp[1-3]
- b. chmod 777 emp
- c. chmod u+r??? emp
- d. chmod 222 emp?

Correct Answer: a

11. Which command is used to remove the directory?

- a. rd
- b. rmdir
- c. dl dir
- d. rdir

Correct Answer: b

12. Find the occurrences of word and print its occurrence count.

Answer: `grep -o -i 'unix' myfile | wc -l`

13. What will be the output of the below command? `find -type f -perm 0777 -print`

- a. Display all files who has all the three permission for all levels of users
- b. Display all directories who has all the three permission for all levels of users
- c. Display all files who has minimum level of permission for all levels of users
- d. Display all folders who has all the three permission for all levels of users

Correct answer: a

14. Which awk script will give you the average of second column of the file named file1, which has '|' as the delimiter?

- a. `awk 'BEGIN{FS=":";s=0}{s=s+$2} END{print s/NR}' file1`
- b. `awk 'BEGIN{FS=":";s=0}{s=s+$2}{print s/NR}' file`
- c. `awk 'BEGIN{FS="|";s=0}{s=s+$2}END{print s}' file1`
- d. `awk 'BEGIN{FS=":";s=0}{s=s+$2}END{print s/NR}' file1`

Correct answer: c

15. Select the correct command to search 'Hello Unix' in a file ignoring case?

- a. `grep 'Hello Unix' file name`
- b. `grep -i 'Hello Unix' file name`
- c. `grep -c 'Hello Unix' file name`
- d. `grep -i file name 'Hello Unix'`

Correct answer: b

16. Which of the following command in unix used to remove the duplicate lines?

- a. `sort -u`

- b. sort uniq
- c. uniq
- d. unique

Correct answer: a

Section 2: TCS ASPIRE – Unix Assignment 1 Quiz (2012)

Q1. Write a command to list all the files inside a folder (including sub-folder files recursively).

Answer: ls -alR

Hint: ls = list information about files. -a = hidden files also. -R = recursive (list subdirectories).

Q2. Search all the files which contain a particular string, say 'include' within a folder.

Answer: grep include /*

Hint: Grep '.' searches named input file for lines containing a match to the given pattern. /* = In current directory (.), all files(*).

Q3. Rename all the files within a folder with suffix 'Unix_'. E.g. if a folder has files a.txt and b.pdf, both should be renamed from a single command to Unix_a.txt and Unix_b.pdf.

Answer: for i in *; do mv "\$i" "Unix_\$i"; done;

Hint: Loop; Syntax for variable in no of times; Do command; Done.

Q4. Rename all files within a folder by the name of the first word of their content (all files should be text files). For example if a .txt contains 'Unix is an OS' in its first line then the file should be renamed to Unix.txt.

Answer: for i in *.txt; do head -1 "\$i" > ad.txt; x=\$(cut -d ' ' -f1 ad.txt); mv "\$i" "\$x.txt"; done;

Hint: i = *.txt (all text files); do head -1 "\$i" > ad.txt; prints first line and store in ad.txt file.

Q5. Suppose you have a C project in a folder called 'project'. It contains .c and .h files and some .txt files and .pdf files. Write a Linux command that will count the number of lines of your text files (total line count of every file, count lines in .txt files only).

Answer: wc -l *.txt

Hint: wc = word count; -l = to count lines; *.txt = all text files.

Q6. Rename all files which contain the sub-string 'foo', replacing it with 'bar' within a given folder.

Answer: for i in *foo*; do mv "\$i" "\${i/foo/bar}"; done

Q7. Show the most commonly used commands from 'history'. (Hint: remember the history command, use cut, and sort it.)

Answer: history|sort -d -1 -f1

Section 3: TCS ASPIRE – Unix Assignment 2 Quiz (2012)

Q1. Create a tree structure named 'training' with 3 sub-directories – 'level 1', 'level2' and 'cep'. Each one further divided into 3. The 'level 1' is divided into 'sdp', 're' and 'se'. How can one reach the home directory in one step, and navigate from 'se' to 'sdp' in one step?

Answer: To navigate from 'se' to home directory in one step: cd To navigate from 'se' to 'sdp' in one step: cd ../b

Q2. How will you copy a directory structure dir1 to dir2 (with all the subdirectories)?

Answer: cp -rlt dir1 dir2

Q3. How can you find out if you have the permission to send a message?

Answer: ls -l (To check access permissions)

Q4. Find the space occupied (in Bytes) by the /home directory including all its subdirectories.

Answer: `du -s -b`

Q5. What is the command for printing the current time in 24-hour format?

Answer: `date +%T`

Q6. Create the following files: chapa, chapb, chapc, chapd, chape, chapA, chapB, chapC, chapD, chapE, chap01, chap02, chap03, chap04, chap05, chap11, chap12, chap13, chap14, and chap15.

Answer: `cat > filename (type contents, then Ctrl+D)`

Example: `cat > chapa` → type contents → Ctrl+D

Q7. What is the command for printing the year, month, and date with a horizontal tab between fields?

Answer: `date +%Y%t%B%t%e`

Q8. What is the command for listing all files ending in small letters?

Answer: `find -name '*[a-z].'`

Q9. What is the command for listing all files ending in capitals?

Answer: `find -name '*[A-Z].'`

Q10. What is the command for listing all files whose last but one character is 0?

Answer: `find -name '*0?.'`

Q11. What is the command for listing all files which end in small letters but not 'a' and 'c'?

Answer: `find -name '*[b-d].'`

Q12. In an organisation, one wants to know how many programmers there are. The employee data is stored in a file called 'personnel'. How can grep be used?

Answer: `grep -c 'programmers' personnel`

Q13. In the organisation mentioned above, how can sed be used to print only the records of all employees who are programmers?

Answer: `sed -n /programmers/p personnel`

Q14. How can sed be used to change the designation 'programmer' to 'software professional' every where in the 'personnel' file?

Answer: `sed 's/programmer/software professional/g' personnel`

Q15. Find out about the sleep command and start five jobs in the background, each one sleeping for 10 minutes.

Answer: `sleep 10m & sleep 10m & sleep 10m & sleep 10m & sleep 10m &`

Q16. How do you get the status of all the processes running on the system (using what option)?

Answer: `ps -e`

Section 4: Most Used UNIX/Linux Commands – Reference

File & Directory Management

Command	Description
ls	Lists files and directories in the current location.
cd	Changes the current working directory.
pwd	Prints the current working directory path.

mkdir	Creates a new directory.
rmdir	Removes an empty directory.
rm	Deletes files or directories (use -r for directories).
cp	Copies files or directories.
mv	Moves or renames files or directories.
touch	Creates a new empty file or updates timestamp.
find	Searches files and directories based on conditions.
locate	Quickly finds files using an indexed database.

File Viewing & Editing

Command	Description
cat	Displays file content.
more / less	Views file content page-by-page.
head	Shows the first few lines of a file.
tail	Shows the last few lines of a file.
nano / vi / vim	Terminal-based text editors.

Compression & Archiving

Command	Description
tar	Archives files into a single .tar file.
gzip / gunzip	Compresses or decompresses .gz files.
zip / unzip	Compresses or extracts .zip files.

System Monitoring & Management

Command	Description
top	Displays running processes and system usage.
htop	Enhanced version of top (if installed).
df -h	Shows disk space usage.
du -h	Shows directory/file size.
free -h	Displays memory usage.
ps	Lists running processes.
kill / killall	Stops a running process.

Networking

Command	Description
ping	Checks network connectivity to a host.
ifconfig / ip addr	Displays network interfaces and IP addresses.

netstat / ss	Shows open ports and connections.
curl	Transfers data from or to a server (e.g., download files).
wget	Downloads files from the internet.

User & Permission Management

Command	Description
adduser / useradd	Adds a new user.
passwd	Changes a user's password.
whoami	Shows the current logged-in user.
sudo	Executes a command with superuser privileges.

File Permissions & Ownership

Command	Description
chmod	Changes file permissions.
chown	Changes file ownership.
umask	Sets default permission for new files.

Miscellaneous

Command	Description
echo	Prints text to the terminal.
date	Displays or sets the system date and time.
history	Shows command history.
alias	Creates a shortcut for a command.
clear	Clears the terminal screen.

Octal File Permissions Reference

Octal Value	Mode	Octal Value	Mode
0	- - -	4	r - -
1	- - x	5	r - x
2	- w -	6	r w -
3	- w x	7	r w x

Octal Value	Symbolic Value	Result
777	a+rx	rxwxrwx
755	u+rx, g=rx, o=rx	rxrx-rx-x
644	u=rw, g=r, o=r	rw-r--r--
700	u=rx, g-rwx, o-rwx	rxw-----

Section 5: Example Commands – grep, sort, awk, sed

grep (search text and patterns)

```
grep 'the' filename
grep -n 'the' filename (adds line count and displays words with 'the')
grep -i 'the' filename (case insensitive)
grep -v 'the' filename (except)
grep -E '[abcd]' filename (display all occurrences of [abcd])
grep -E '\w{6,}' filename (display words having >= 6 letters)
```

Pipes to connect commands (|)

```
echo 'hello, dinesh' | wc (displays no. of lines, words, chars)
```

View text files with cat, head, tail and less

```
cat filename (displays content in file)
head filename (displays first 10 lines)
tail filename (displays last 10 lines)
cat -n filename (add line count)
cat filename | tail filename | cat -n
head -n5 filename (displays 5 lines)
tail -n5 filename (displays last 5 lines)
```

Manipulate text with awk, sed and sort

```
awk '{print $2 "\t" $1}' filename (displays 2nd col and 1st col)
awk '{print $2 "\t" $1}' filename | sort -n (sorted with numeric values)
sed 's/orange/blue/' filename (changes the string orange to blue in the file)
```

SORT

```
sort filename (sorts according to first char of 1st col)
sort -k2 filename (sorts according to first char of sec col/key)
sort -k2 -n filename (sorts according to numeric values of sec col)
sort -u filename (displays only unique lines / removes duplicates)
```

rev

Print text in reverse sequence.

tac

Concatenate and print files in reverse.

tr

Translate or modify individual characters according to parameters.

Creating tar and zip archives

```
tar cvf filename.tar filestobearchived (makes a tar file)
tar caf filename.tar.gz filestobearchived (makes a compressed zip)
tar caf filename.tar.bz2 filestobearchived (makes a compressed zip)
tar xf filename.tar.gz/bz2 -C foldername (unpacks into folder)
zip filename.zip contentfilename
unzip filename.zip
```